

Supplement A: Mathematical Equivalence of Clinical Calibration Across Within-Effect-Modifier Normalizers

This supplement establishes that the clinical-scale quantity $\Delta_{\max}$, used as the calibration output of the methodology in the main paper, does not depend on the choice of within-effect-modifier scale normalizer applied to the Wasserstein-1 distance. This invariance result is referenced at Section 2.2 (Per-EM Wasserstein-1 Distance) and Section 2.4 (Interpretation and Clinical Calibration) of the main paper and provides the mathematical basis for our adoption of W_1 in its original measurement units as the primary similarity metric.

1 Setup and Definitions

Let F_1 and F_2 denote probability distributions on $\mathbb{R}$ representing the marginal distribution of an effect modifier in two regions of a multi-regional clinical trial. Let $\bar{F}$ denote a pooled distribution constructed from F_1 and F_2 (for example, the equal-weight mixture $\frac{1}{2}F_1 + \frac{1}{2}F_2$ or a sample-size-weighted mixture). Let $W_1(F_1, F_2)$ denote the Wasserstein-1 distance between F_1 and F_2 , expressed in the original measurement units of the effect modifier.

A within-effect-modifier scale functional is any map $X : \mathcal{P}(\mathbb{R}) \rightarrow \mathbb{R}_+$ that returns a strictly positive scalar carrying the units of the input distribution. Common choices considered in the prior literature on distributional similarity include

- the interquartile range $X_{\text{IQR}}(F) = F^{-1}(0.75) - F^{-1}(0.25)$,
- the standard deviation $X_{\text{SD}}(F) = \sqrt{\text{Var}_F(X)}$,
- the median absolute deviation about the median, $X_{\text{MAD}}(F)$,
- the inter-decile range $X_{\text{Q95-Q5}}(F) = F^{-1}(0.95) - F^{-1}(0.05)$,
- the range $X_{\text{rng}}(F)$ (when finite).

For any such functional with $X(\bar{F}) > 0$, a dimensionless normalized similarity index can be defined as

$$\text{nABCD}_X(F_1, F_2) = \frac{W_1(F_1, F_2)}{X(\bar{F})}. \quad (1)$$

The clinical calibration formula in the main paper (Section 2.4, equation for $\Delta_{\max}$) translates a similarity index into a maximum potential regional treatment effect difference. Expressed via a normalizer-dependent index, this calibration takes the form

$$\Delta_{\max}^{(X)} = L_{\text{clinical}} \cdot X(\bar{F}) \cdot \text{nABCD}_X(F_1, F_2), \quad (2)$$

where L_{clinical} is the Lipschitz constant of the conditional average treatment effect function with respect to the effect modifier (the clinical slope).

2 Equivalence Result

Proposition 1. *For any within-effect-modifier scale functional X such that $X(\bar{F}) > 0$,*

$$\Delta_{\max}^{(X)} = L_{\text{clinical}} \cdot W_1(F_1, F_2), \quad (3)$$

and the right-hand side of equation (3) does not depend on the choice of X .

Proof. Substituting equation (1) into equation (2),

$$\Delta_{\max}^{(X)} = L_{\text{clinical}} \cdot X(\bar{F}) \cdot \frac{W_1(F_1, F_2)}{X(\bar{F})} = L_{\text{clinical}} \cdot W_1(F_1, F_2).$$

The two factors of $X(\bar{F})$ cancel identically. Because the right-hand side contains no reference to X , the value of $\Delta_{\max}^{(X)}$ is the same for every admissible choice of X . $\square$

Remark 1 (Estimator-level invariance). *The invariance extends to plug-in estimators. When W_1 is estimated by $\widehat{W}_1$ and the normalizer is estimated by $\widehat{X(\bar{F})}$ from the same sample, the corresponding estimator of $\Delta_{\max}^{(X)}$ is*

$$\widehat{\Delta_{\max}^{(X)}} = L_{\text{clinical}} \cdot \widehat{X(\bar{F})} \cdot \frac{\widehat{W}_1}{\widehat{X(\bar{F})}} = L_{\text{clinical}} \cdot \widehat{W}_1.$$

Consequently, the nonparametric bootstrap distribution of $\widehat{\Delta_{\max}^{(X)}}$ coincides with that of $L_{\text{clinical}} \cdot \widehat{W}_1$ for every choice of X . Percentile bootstrap confidence intervals for $\Delta_{\max}$ are therefore numerically identical regardless of whether one routes the computation through a normalized intermediate quantity or directly through raw $\widehat{W}_1$.

3 Connection to the Reverse Pathway

The reverse calculation introduced in the main paper (Section 2.4) defines the required clinical slope as $L^* = \Delta_{\text{clin}}/W_1(F_1, F_2)$, where Δ_{clin} is a pre-specified clinical margin. Substituting any normalized form via $W_1 = X(\bar{F}) \cdot \text{nABCD}_X$ yields

$$L^* = \frac{\Delta_{\text{clin}}}{X(\bar{F}) \cdot \text{nABCD}_X(F_1, F_2)},$$

which simplifies to the same $L^* = \Delta_{\text{clin}}/W_1$. The reverse pathway is therefore invariant to normalizer choice on the same grounds as the forward pathway.

4 Implications for Methodology Choice

Proposition 1 and Remark 1 together establish three operational consequences for methodology design:

1. Adopting W_1 in its original measurement units produces the same clinical calibration as adopting any normalized form nABCD_X followed by multiplication by $X(\bar{F})$. There is no methodological loss from routing the calibration directly through raw W_1 .

2. The choice of X (interquartile range, standard deviation, median absolute deviation, and so on) affects only the value of an intermediate dimensionless display quantity. Empirical sensitivity studies comparing $nABCD_X$ across different choices of X measure the sensitivity of the intermediate display, not of the clinical conclusion.
3. Because the normalizer choice is mathematically inert at the clinical-scale layer but introduces an arbitrary scientific convention at the dimensionless layer, the principle of statistical appropriateness favors adopting W_1 in its original measurement units as the primary similarity metric. Within-effect-modifier normalization is unnecessary once clinical calibration is applied.

The main paper adopts this approach throughout. Where prior literature has introduced normalized variants of W_1 for cross-effect-modifier comparison, the present supplement clarifies that such normalization is unnecessary within the per-effect-modifier methodology presented in the main text. Cross-effect-modifier comparison, which would require navigating the dimensional heterogeneity across variables measured in different units, is explicitly outside the scope of the methodology (Section 5 Discussion of the main paper).